Heuristics and trade-offs

Optimization · Level 4 · Lesson 4



In this lesson you will

- Explain NP-hardness in plain words and why heuristics are used.
- Implement 2-opt and compare metaheuristics for escaping local optima.
- Measure the optimality gap and put fairness limits in the open.

Key ideas

- Build fast, then improve with local search
- Escape local optima: annealing, tabu, genetic
- Gap = (found - bound) ÷ found
- Several goals: show the trade-off with numbers

Activity: Route race

1. Build (6 min): Nearest neighbour on a paper map.
2. Improve (8 min): Uncross with 2-opt moves.
3. Check (6 min): Compare with the best route.
4. Fair (5 min): Add a limit on the longest ride.

Use a real local map if you can; travel times work as well as distances. Work in pairs and swap routes to check each other’s sums.

Let’s discuss

1. Why do companies accept routes nobody has proved optimal?
2. Which metaheuristic would you try first for a timetable, and why?
3. Who should decide the longest acceptable ride on a school bus?

Check yourself

- Your route is 52 km and a bound says 50 km. Write the gap, and one sentence a manager would understand.